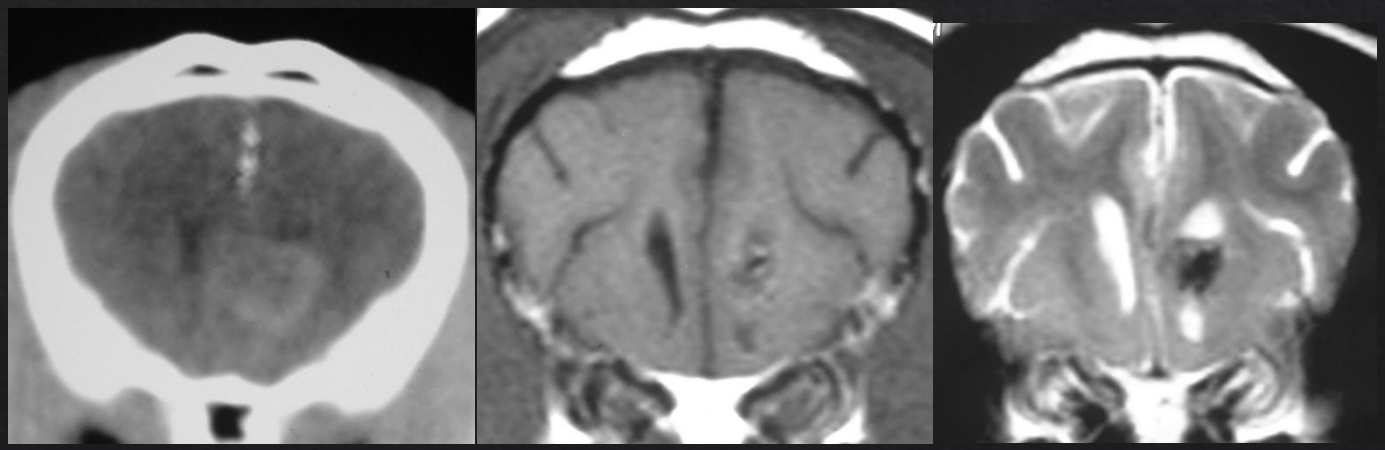
CT and MRI

Clinical Applications in Veterinary Medicine

張雅珮 BVM, MVM, PhD, DipECVN, DAiCVIM (Neurology)



CT and MRI

- Cross sectional images
- Digital processing
- Sedation / anesthesia required in animals
- Comparison to radiography / ultrasonography

CT vs. MRI – the differences

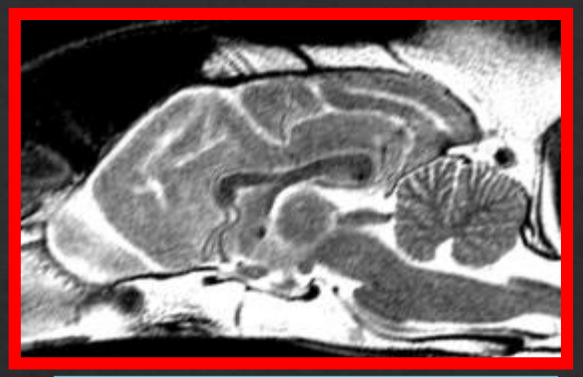
- Principles
- Imaging quality in different structures
- Image plane

CT

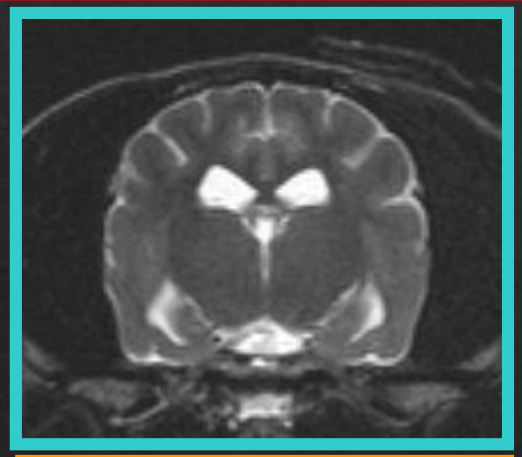
- Transverse planes
- Reconstruction

MRI

- Any planes



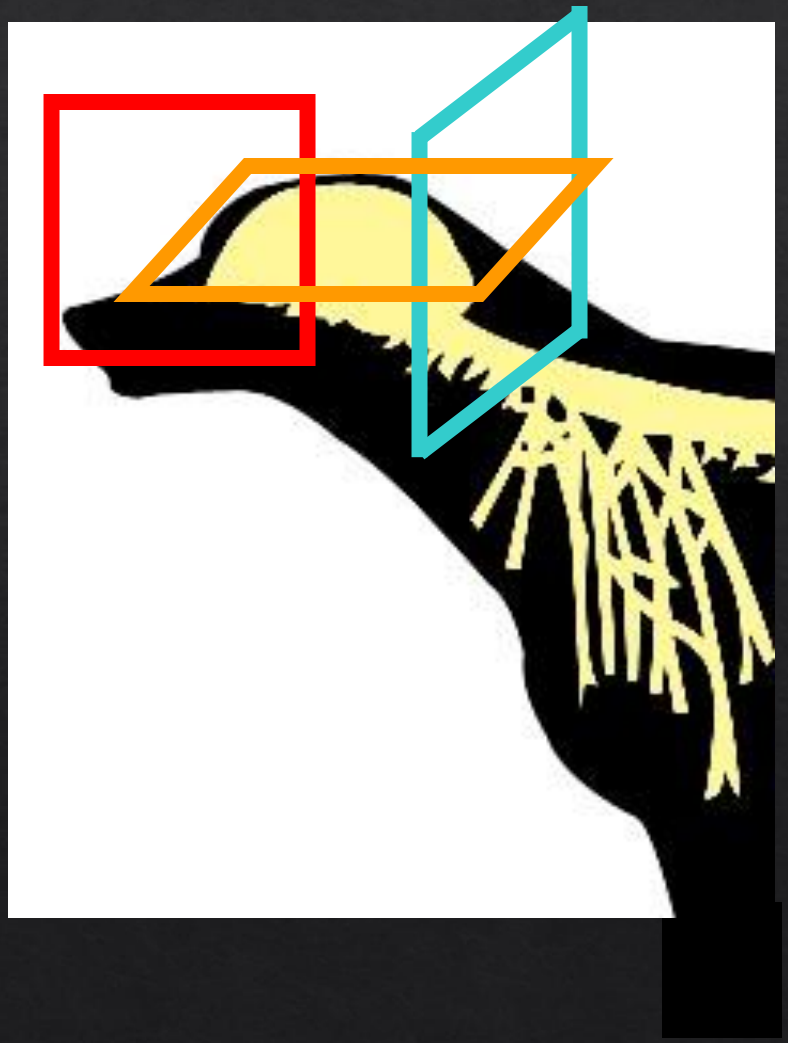
Sagittal



Transverse



Dorsal



CT or MRI ?

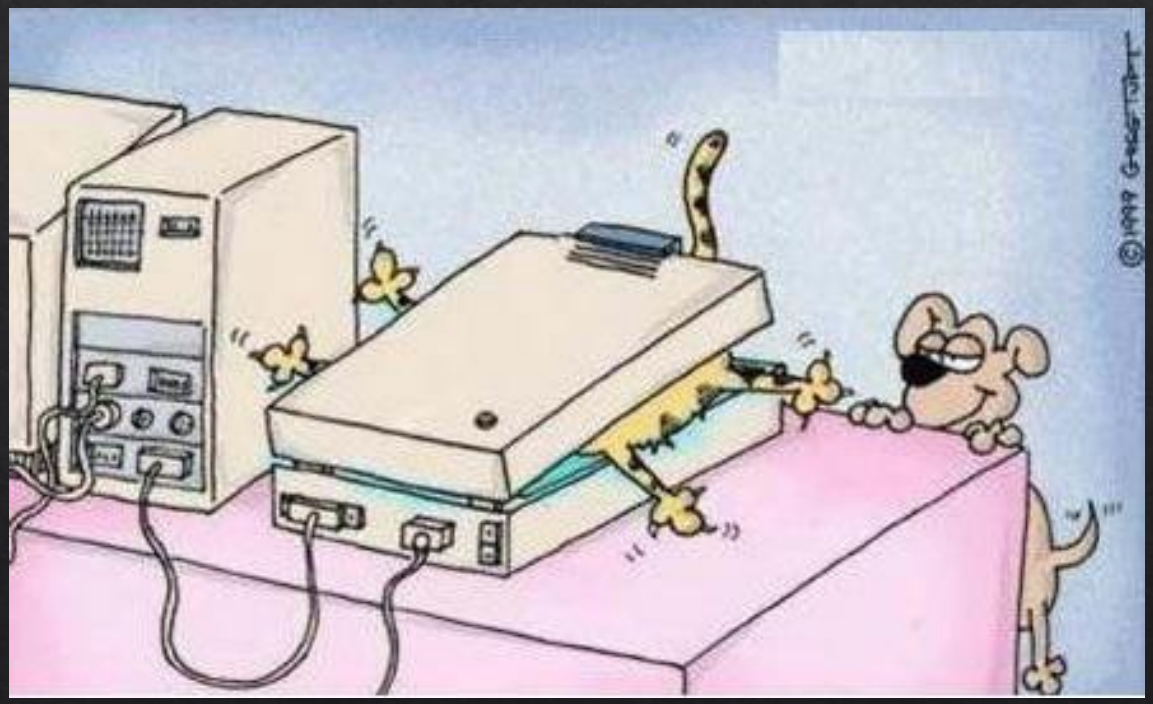
CT	MRI
Familiar with normal anatomical structures	
Hounsfield unit (HU) CT windowing (WW / WL)	Tissue characteristic on different sequence
Bone +++ Soft tissue ++ (lesion shape, contrast enhancement)	Soft tissue ++++
Short anesthesia time	Longer anesthesia time
User friendly	Special training required

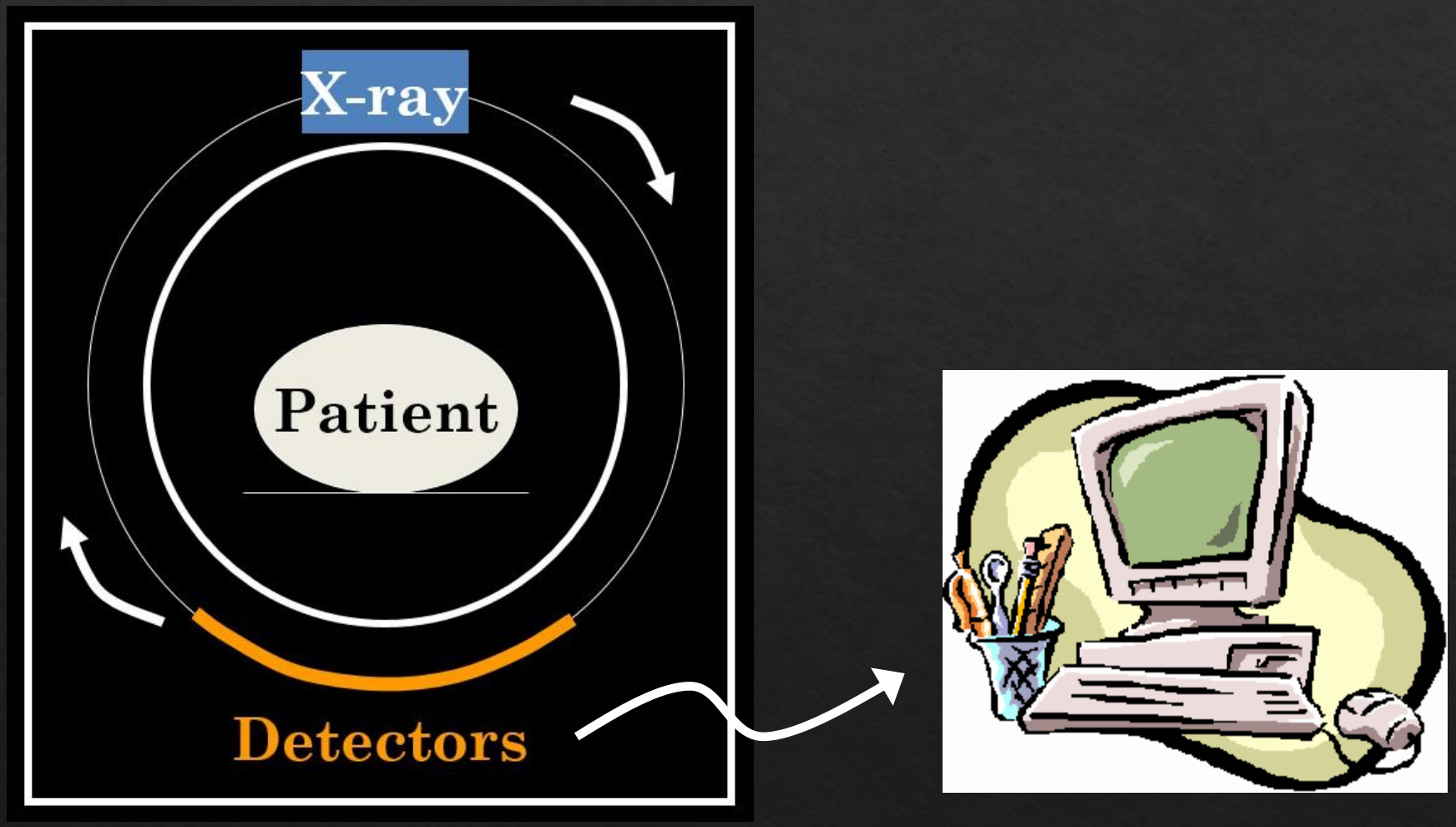
CT or MRI ?

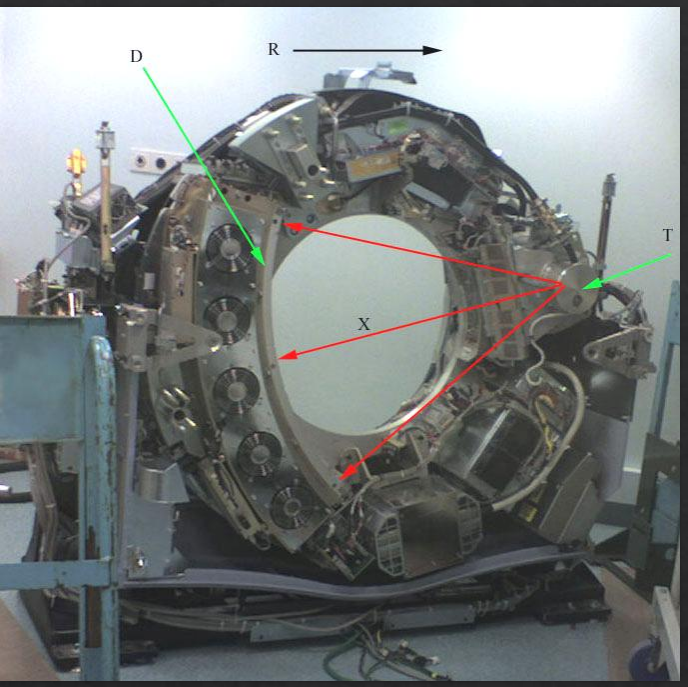
	CT	MRI
Nasal cavity	++	++
Retrobulbar region	++	++
Brain	+	++++
Middle ear	++	++
Skull base structure	++	+++
Muscle	+	+++
Thorax	+++	–
Abdomen	+++	+++ (High-field MRI)
Spinal cord / vertebral canal	+ (IVDD)	+++
Bone / vertebra / skull	++++	+

CT

Computed (axial) tomography




X-ray
Patient
Detectors

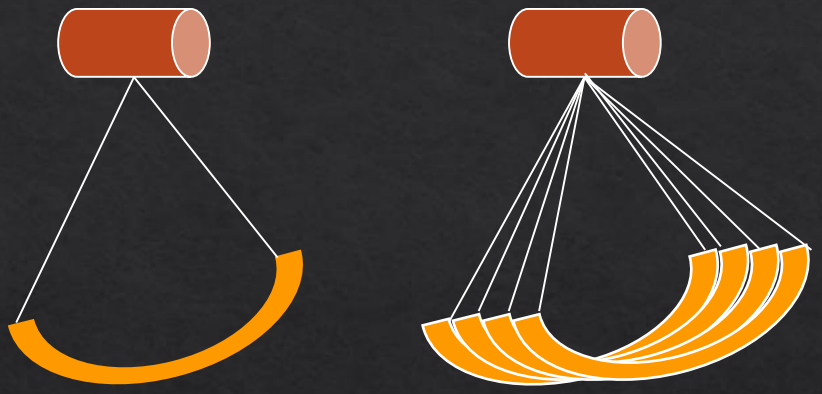

D
R
T
X

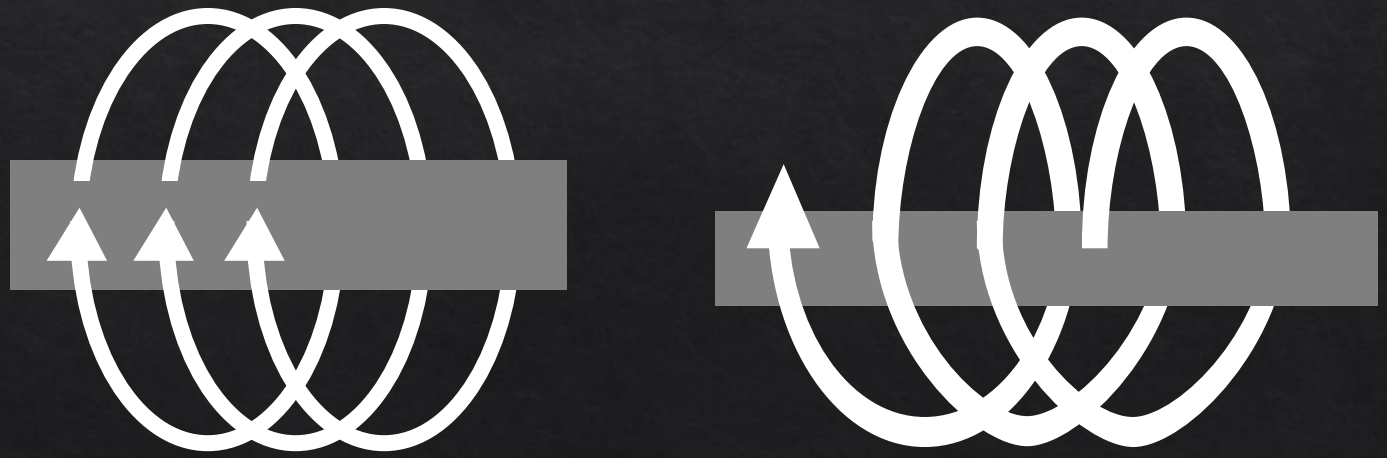

TOSHIBA


TOSHIBA
MALLINCKRODT
TOSHIBA

Types of CT machine

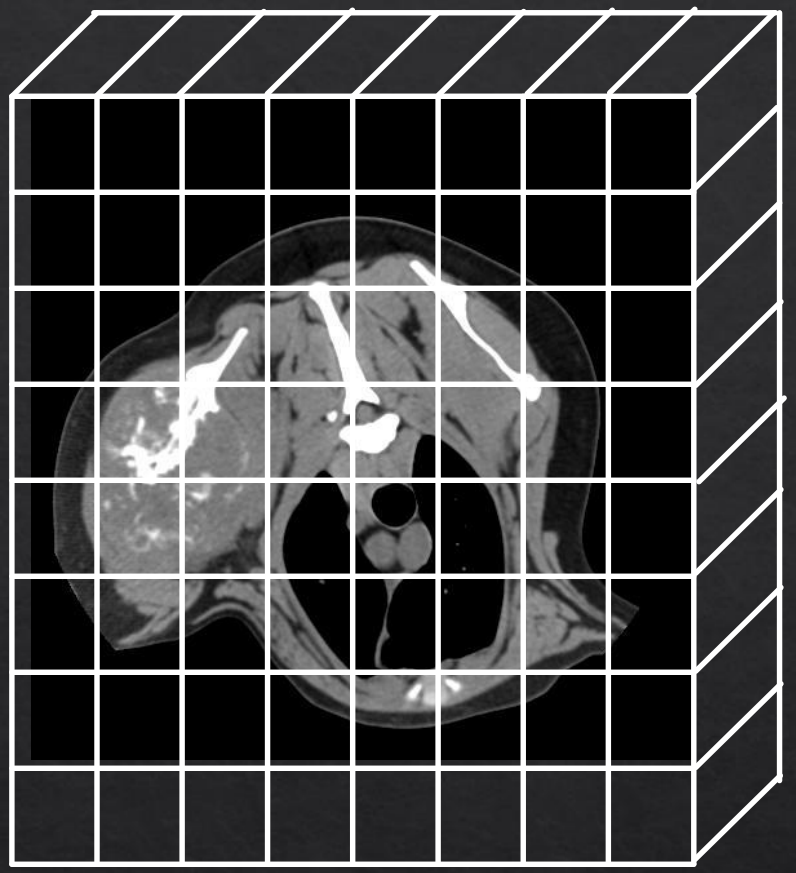
- Single slice CT
- Spiral or helical CT
- Multislice CT







Matrix size = $N_1 \times N_2$





Hounsfield unit (CT number)

Pixel

Voxel
= Pixel × slice thickness

Hounsfield Unit (HU)
CT number

The averaged attenuation values of the voxel

↓

Presented in number

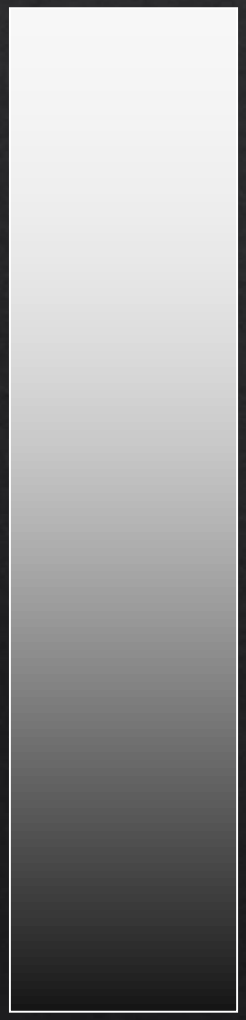
↓

Compared to that of water

↓

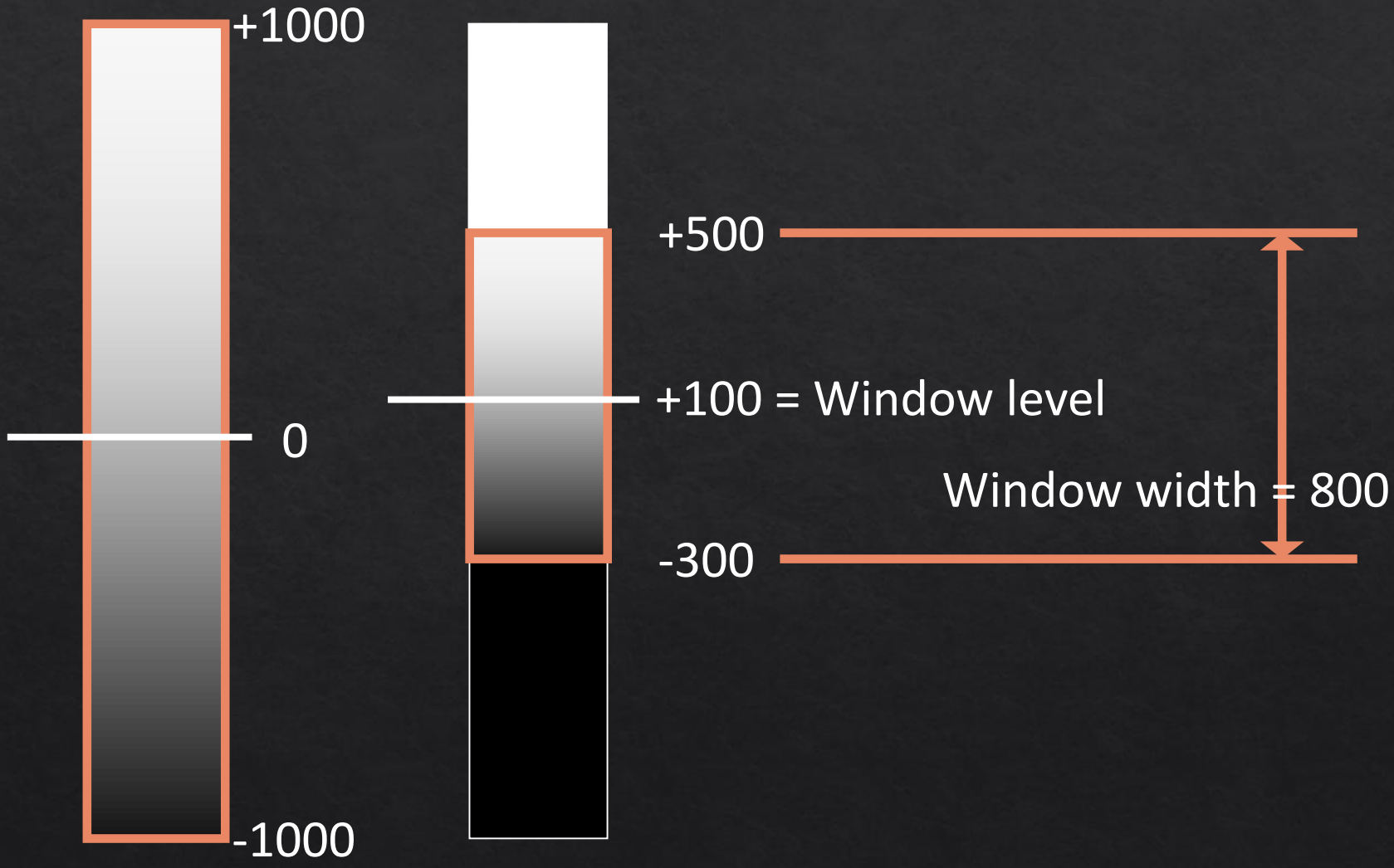
Hounsfield unit (CT number)

Hounsfield Unit (HU)
CT number



Bone	+1000
Clotted blood	+50 to +90
Abdominal organs	+50
Brain	+30
Water	0
Fat	-100
Lung	-845
Air	-1000

CT windowing



CT windowing

Window level (WL) / Window center (WC)

- HU of the tissue of interest

Window width (WW)

- Narrow WW
  - Increase tissue contrast
- Wide WW
  - Good inherent tissue contrast
  - Wide range of tissue needs to be displayed

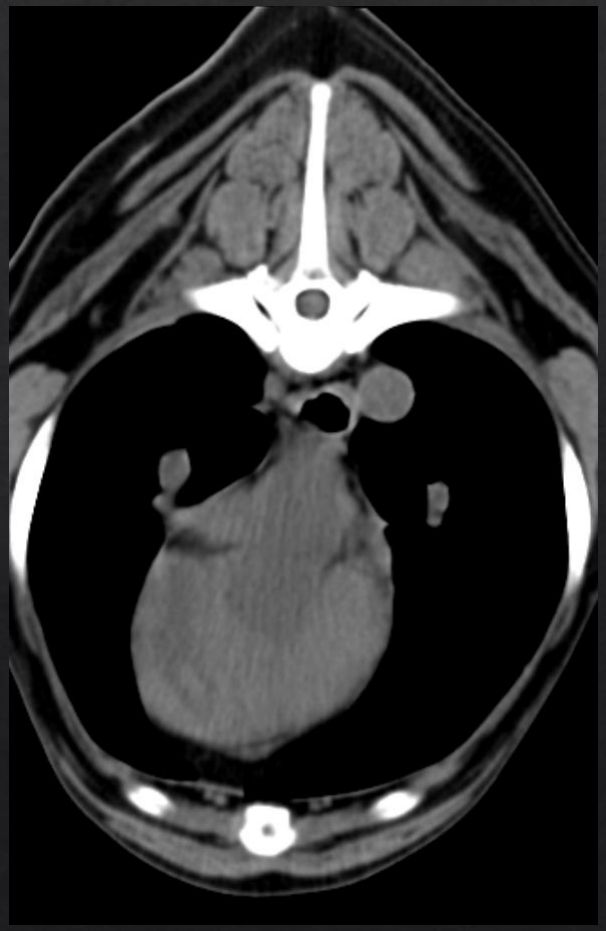
Bone window

WL 300
WW 2000
Range -700 to 1300



Soft tissue window

WL 50
WW 350
Range -125 to 225



Lung window

WL -600
WW 1600
Range -1400 to 200



Sinus window

WL	400
WW	4000
Range	-1600 to 2400

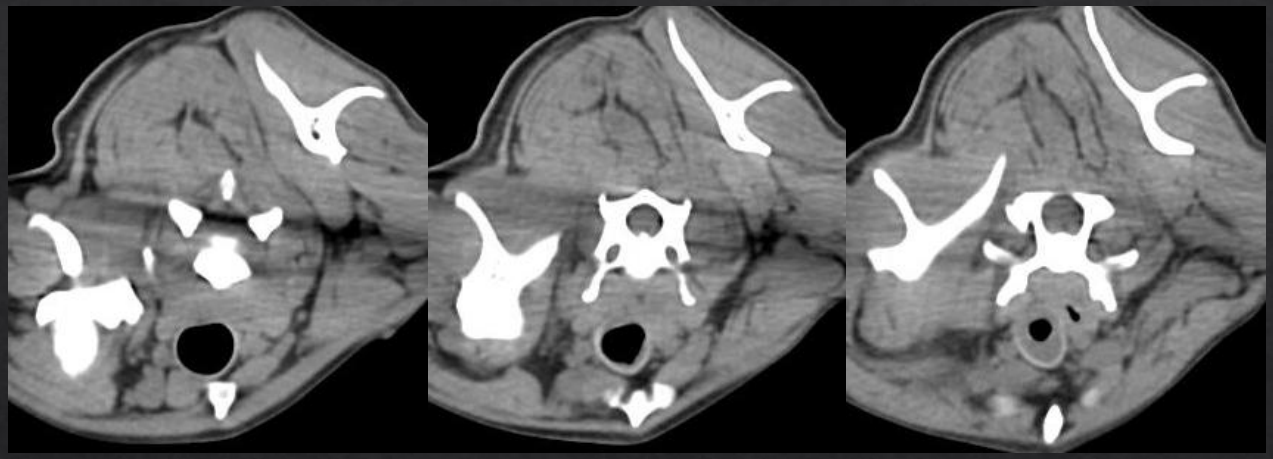


Soft tissue window

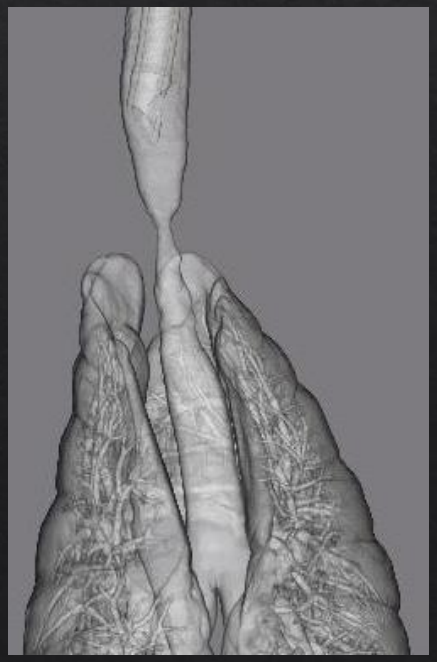
WL	50
WW	350
Range	-125 to 225



Image reconstruction





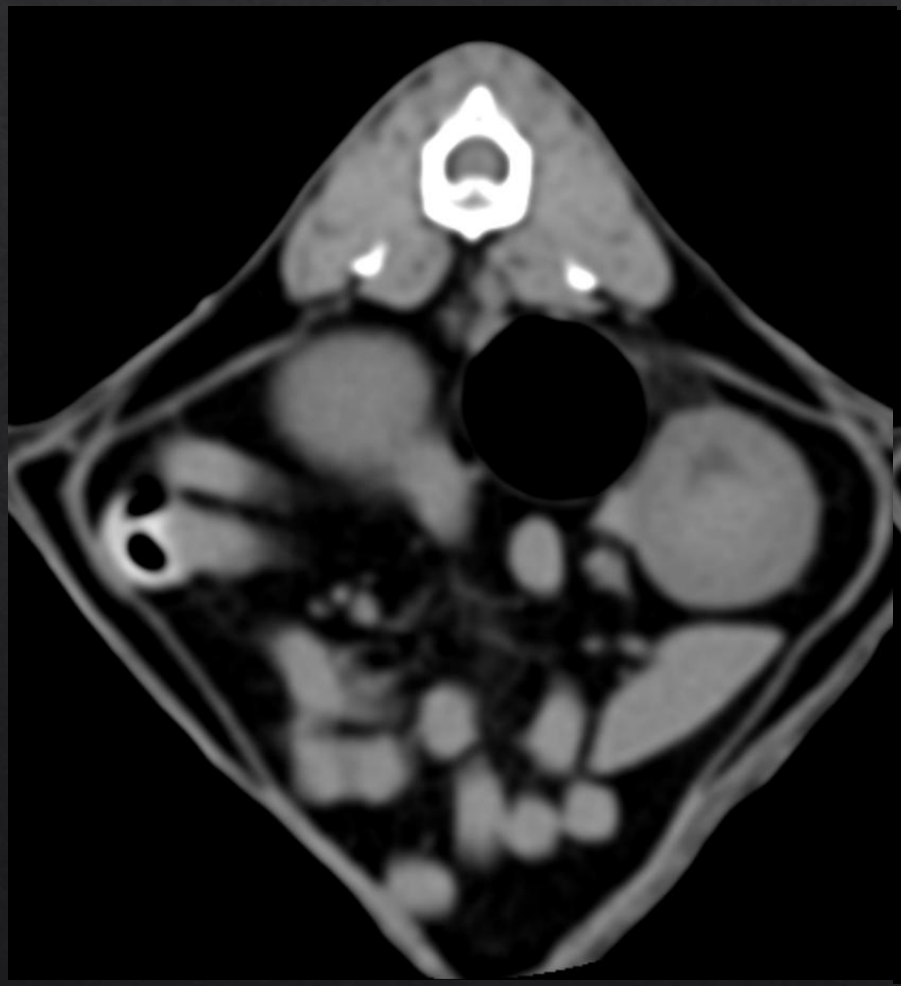


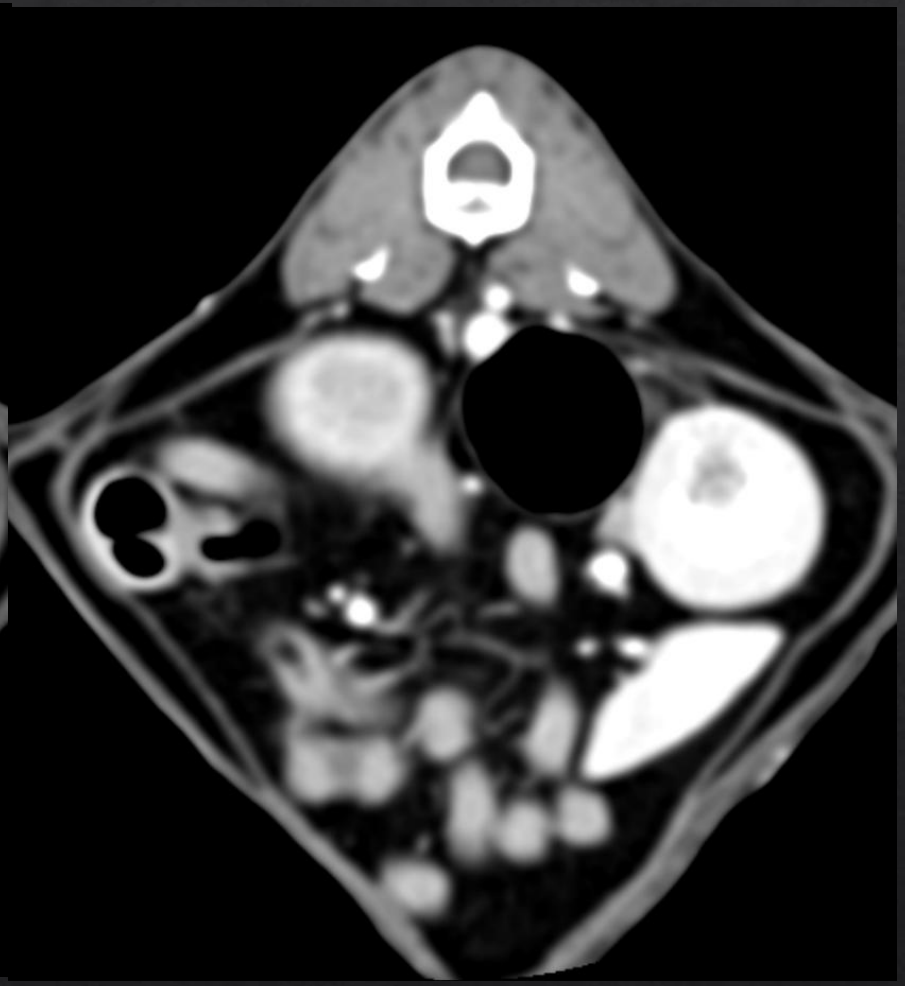
Contrast media

- Iodinated contrast media
- Intravenous
  - Enhancement
    - Normal tissue
    - Lesion
    - CNS: blood-brain-barrier
    - Blood vessels ➔ CT angiography
  - Excretion by glomerular filtration ➔ CT urography

Pre-contrast

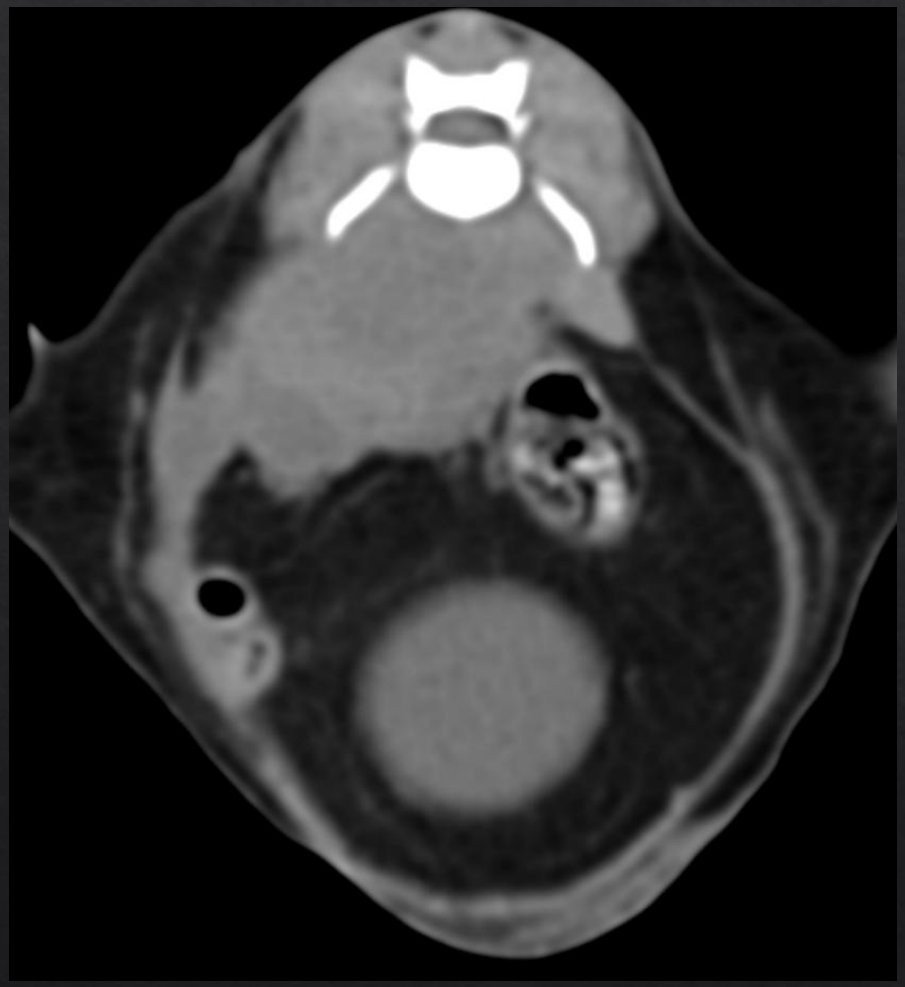
Post-contrast

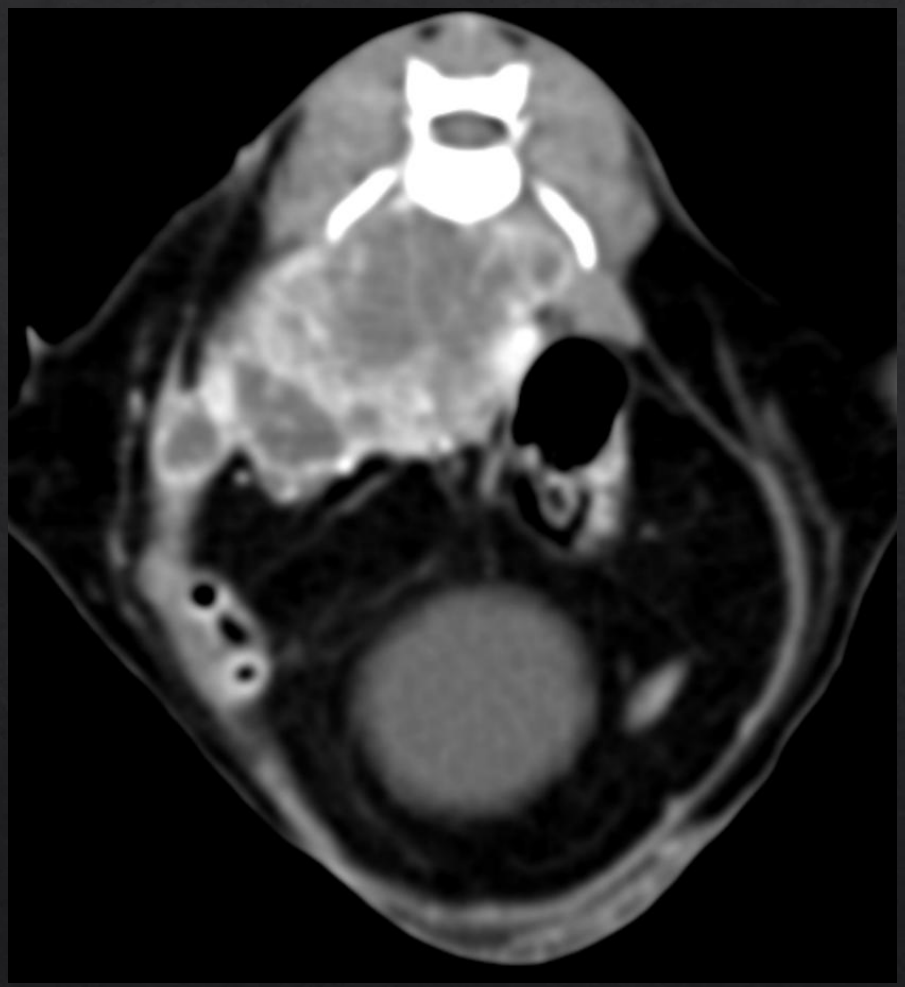




Pre-contrast

Post-contrast





Sub-lumbar mass

Contrast reaction - human

- Low-osmolality contrast media: safer (0.2 %)

Mild	Skin rash, itching, nasal discharge, nausea, vomiting
Moderate	Persistence of mild symptoms, facial or laryngeal edema, bronchospasm, dyspnea, tachycardia, bradycardia
Severe	Life-threatening arrhythmias, hypotension, laryngeal edema, overt bronchospasm, pulmonary edema, seizure, syncope, death

Contrast-induced nephrotoxicity

- Risk factors (human)
  - Pre-existing renal disease
  - Diabetes mellitus
  - Multiple myeloma
  - Proteinuria, dehydration, hypertension

Contrast reactions - Prevention

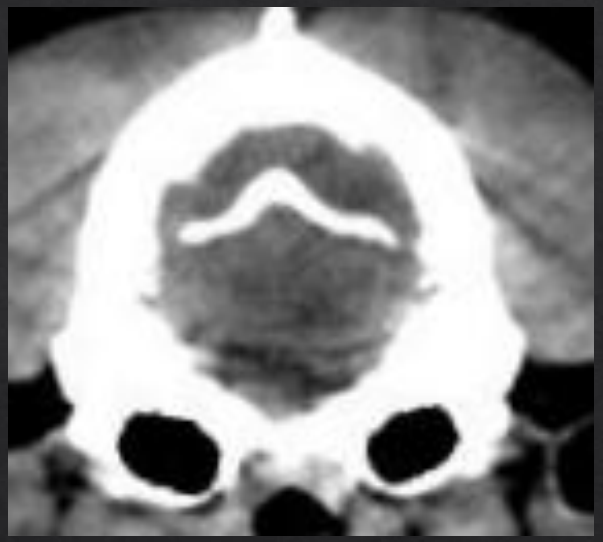
- Is contrast study really required ?
- Use low-osmolality contrast media
- Contrast reaction
  - General anesthesia
  - Premedication in patients with previous reactions
- Nephrotoxicity
  - Check renal function prior to CT study
  - Use minimum effective doses
  - Maintain adequate hydration
  - Stop other nephrotoxic medications before CT study

Artifacts

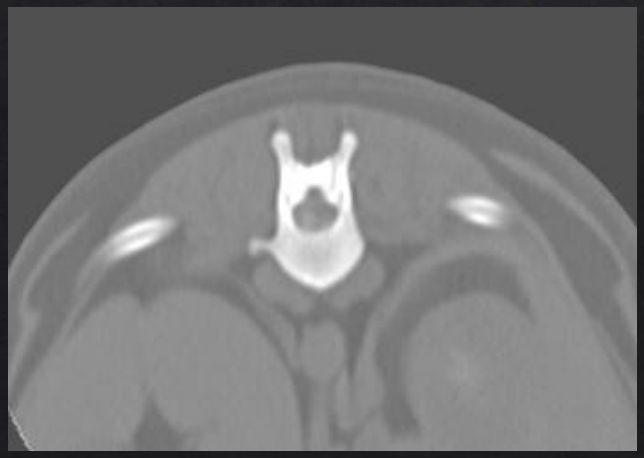
Motion



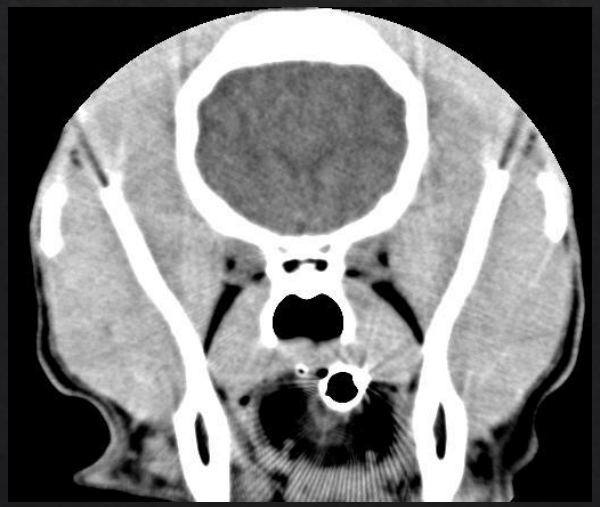
Beam hardening



Partial volume averaging



Others



MRI

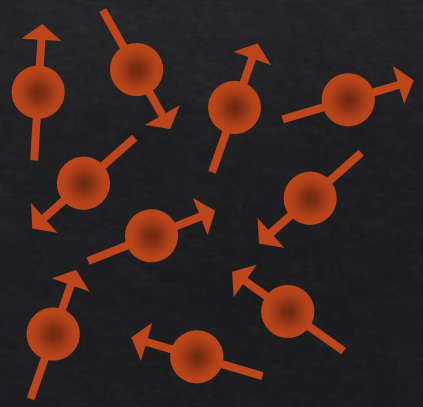
Magnetic resonance imaging

Magnetic resonance imaging

Hydrogen proton

Positive electrical charge → Magnetic field

Spinning about its own axis





$M_0 = 0$

MRI

- Most clinical magnet: 0.5 – 3 tesla (T)
- Low field MR: 0.2 – 0.5 tesla (T)
- High field MR: 1 – 3 T tesla (T)
  - Most human hospitals



Metallic objects in the magnetic field

- Safety issue











- Metal implants

  Devices with magnetic field - cardiac pacemaker, clips, orthopedic implants…

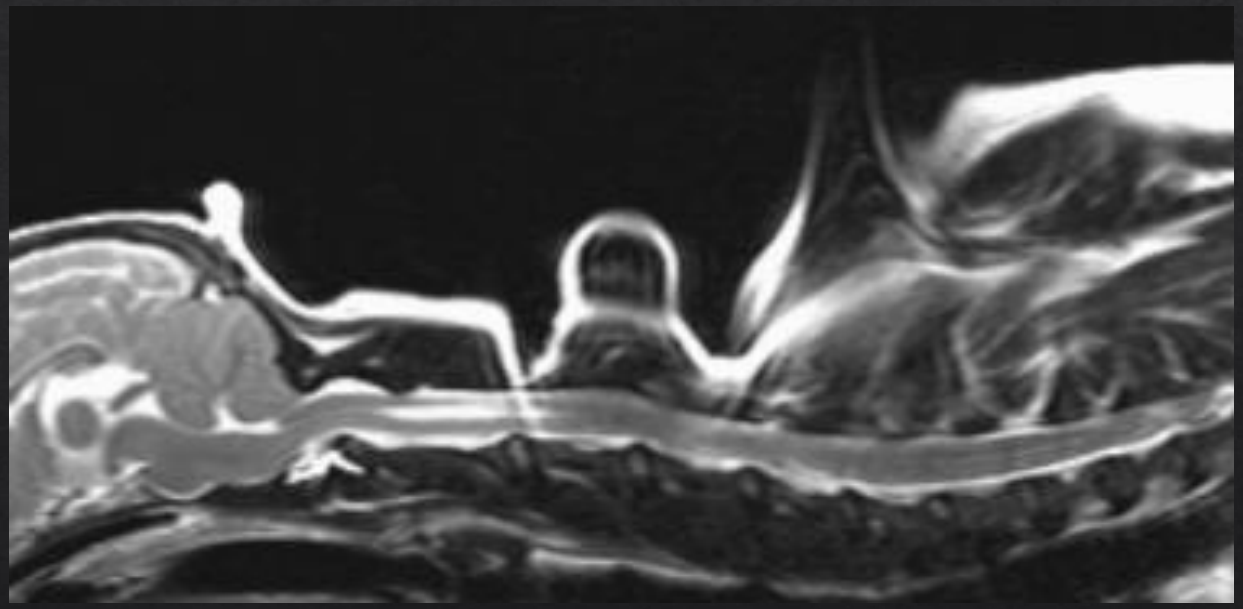
- Wallet, credit card, watch, keys…

  collar, leash…

- Monitoring equipments, needles, stethoscope, trolley...

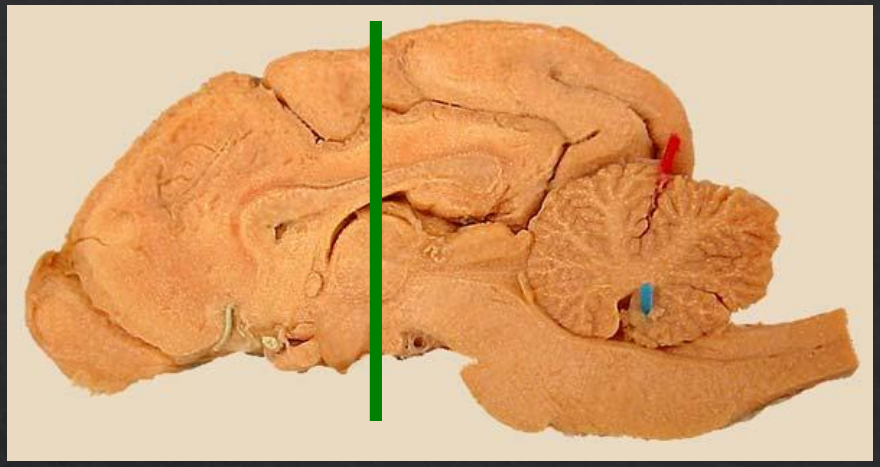
Metallic objects in the magnetic field

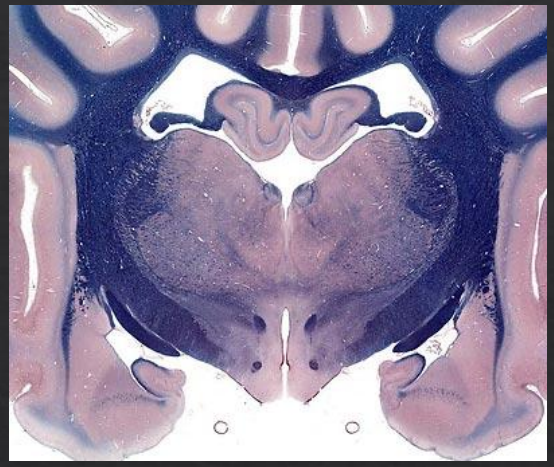
- Image distortion



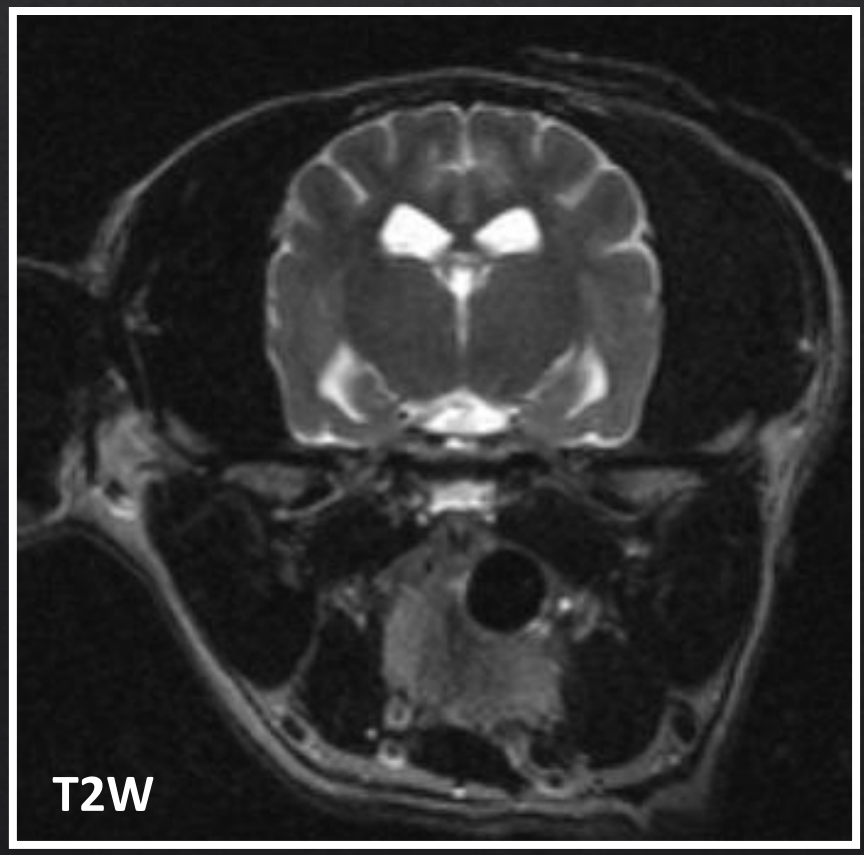
Common pulse sequences in MRI

- T1-weighted
- T2-weighted
- T1-weighted post-gadolinium
- FLAIR (Fluid-attenuated Inversion Recovery)
- GRE (Gradient Echo) / SWI (Susceptibility Weighted Imaging)
- Fat-Signal Suppression
  - STIR (Short T1 Inversion Recovery)
  - Chemical Fat-Saturation



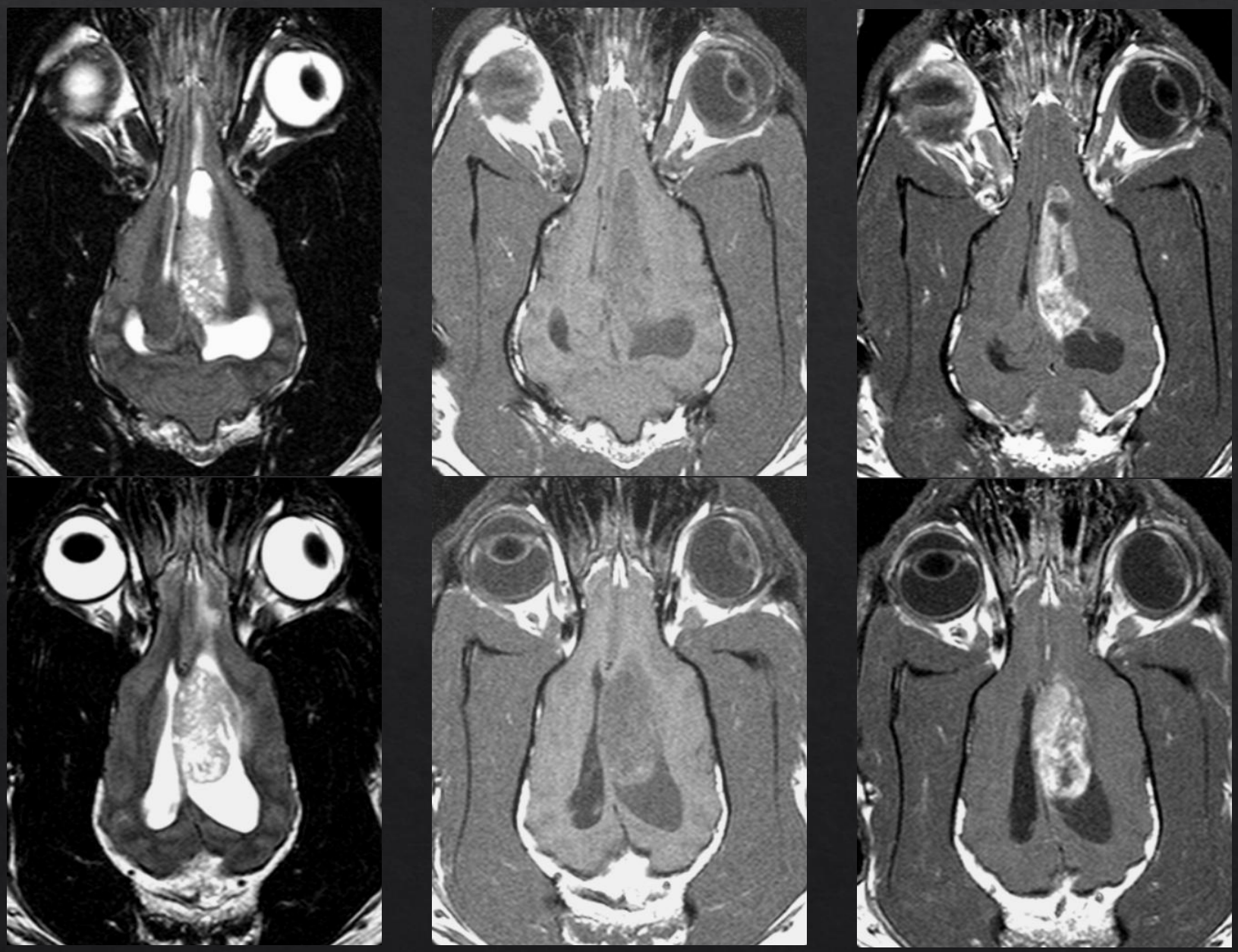



T1W


T2W

Contrast media – Gadolinium (Gd / Gad)

- Paramagnetic property
- Area with contrast enhancement
  - Blood vessels (normal / increased vasculature)
  - Normal tissue
  - Lesion
  - Disruption of blood-brain barrier
- Post-Gd images: T1-weighted, FLAIR
- Contrast reaction: lower than non-ionic iodinated contrast media

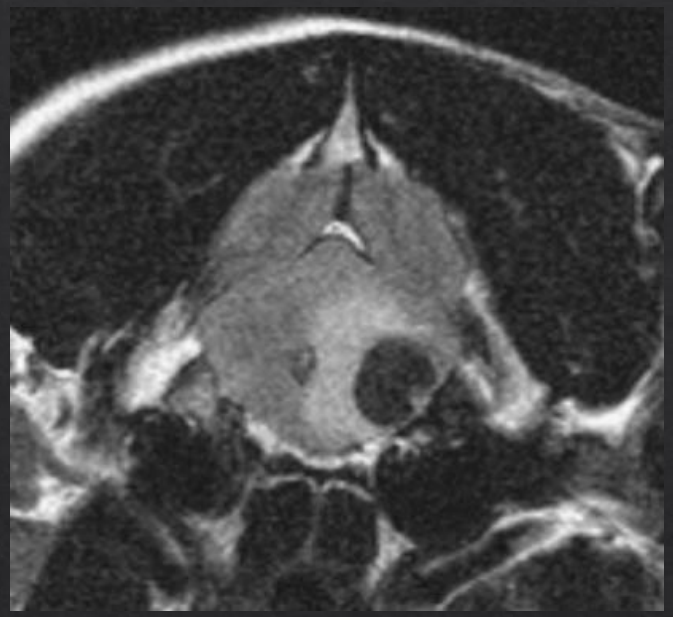

T2W
T1W
T1W Post-Gd

FLAIR
(Fluid-attenuated Inversion Recovery)

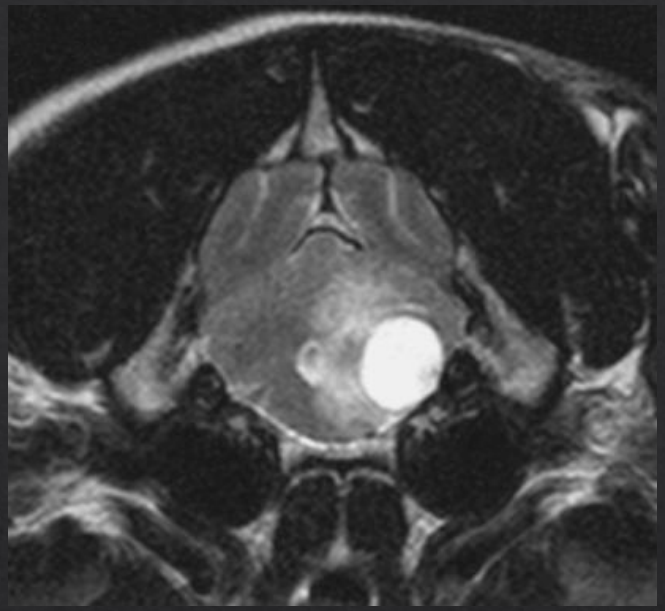
- Elimination of the signal from pure fluid
- Detecting lesions close to fluid-containing space
  - Ventricles
  - Sulci
  - Cyst

Case 1

FLAIR

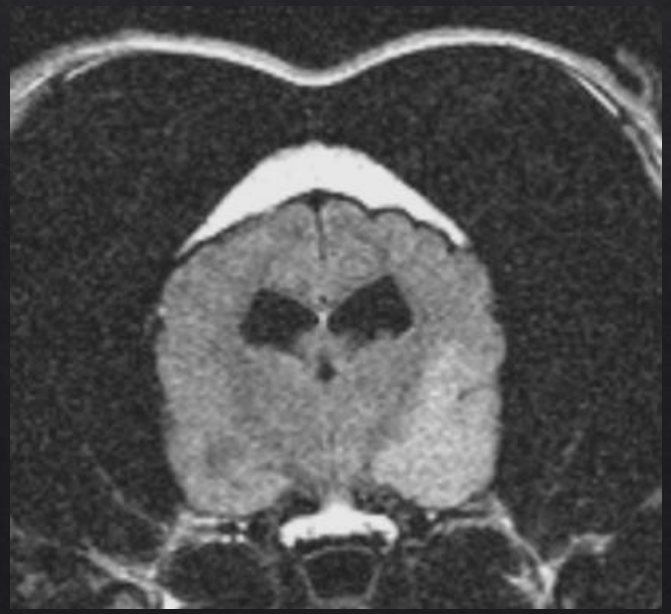


T2W

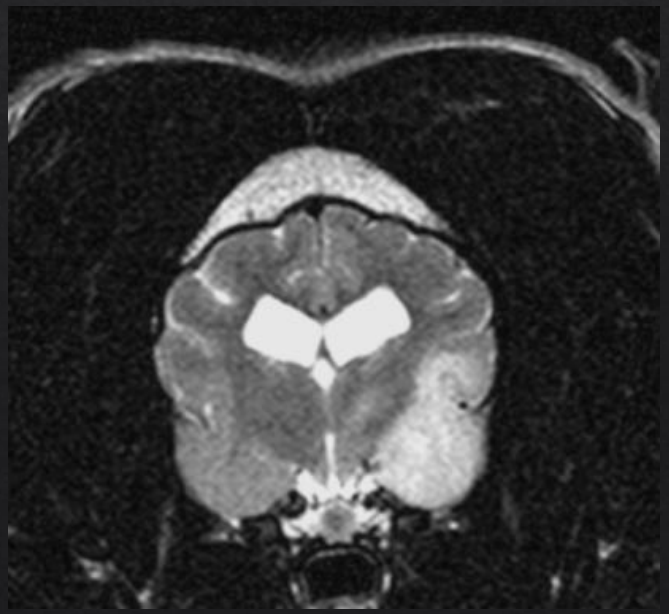


Case 2

FLAIR



T2W



GE (Gradient Echo)

- Sensitive to difference in magnetic susceptibility between tissues

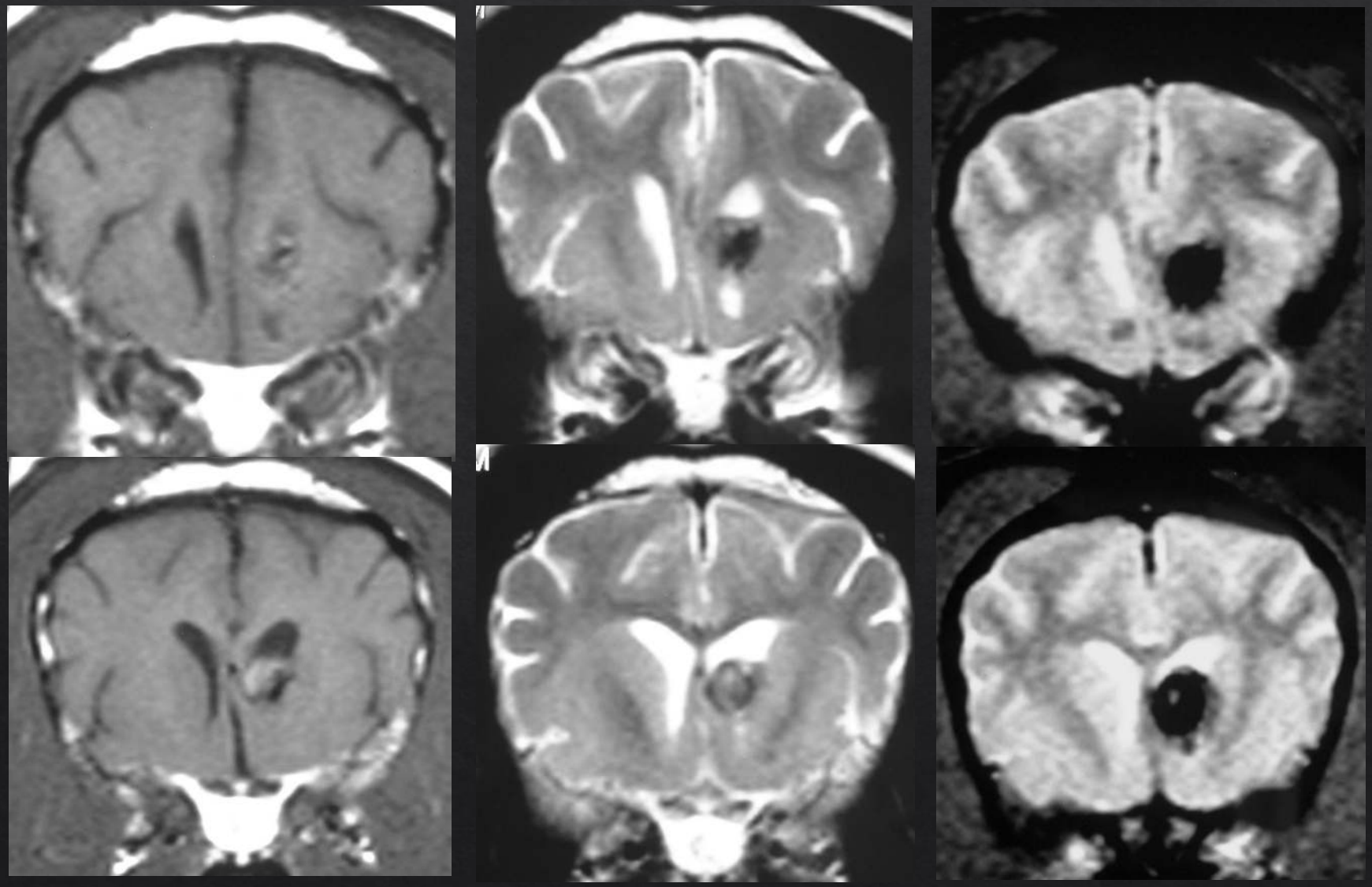
- Paramagnetic or diamagnetic substances

  Methemoglobin, iron (ex. ferritin), copper, melanin, calcification, foreign metallic substances

  ⬇

  Magnetic susceptibility artifacts on GE sequences

  Signal void on GE

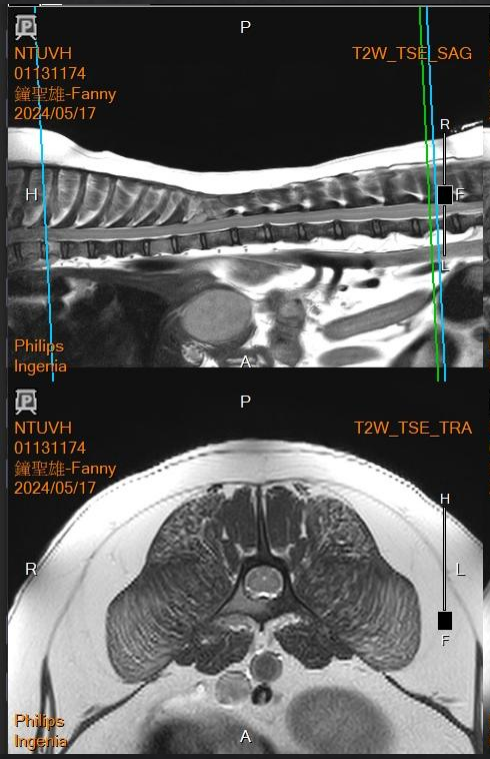

T1W
T2W
GE

Fat-signal suppression techniques

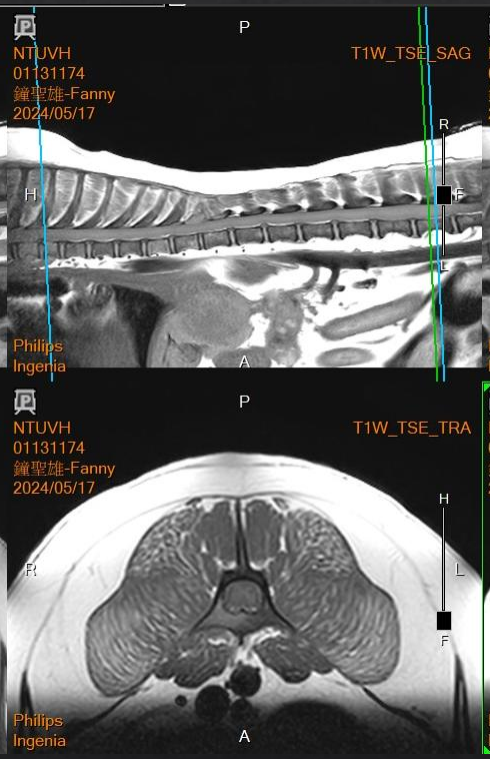
- STIR (short T1 inversion recovery)
- Chemical fat-saturation
- Elimination of the signal from fat
- Detecting lesions close to fat tissue
  - Bone marrow lesion
  - Lipoma
  - Brachial plexus region

L4

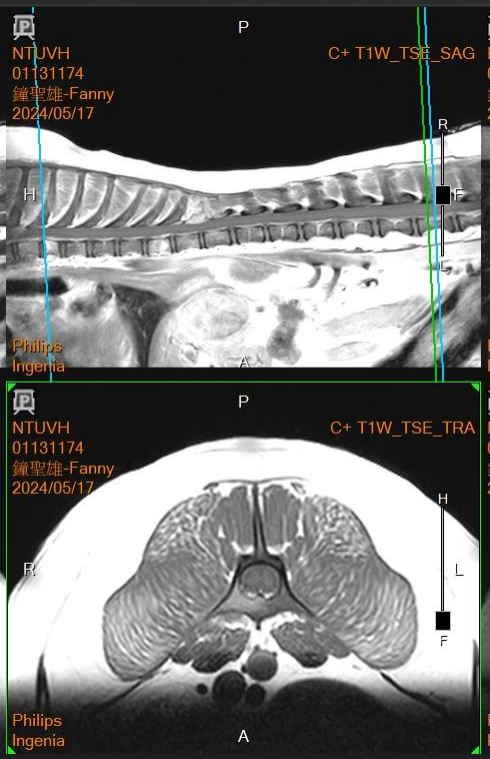
T2W



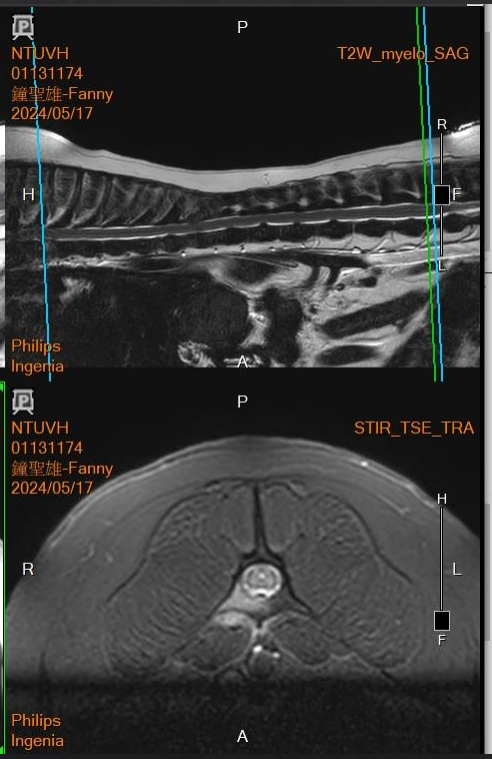
T1W



T1W Post-Gd



STIR



T1W

T2W

STIR

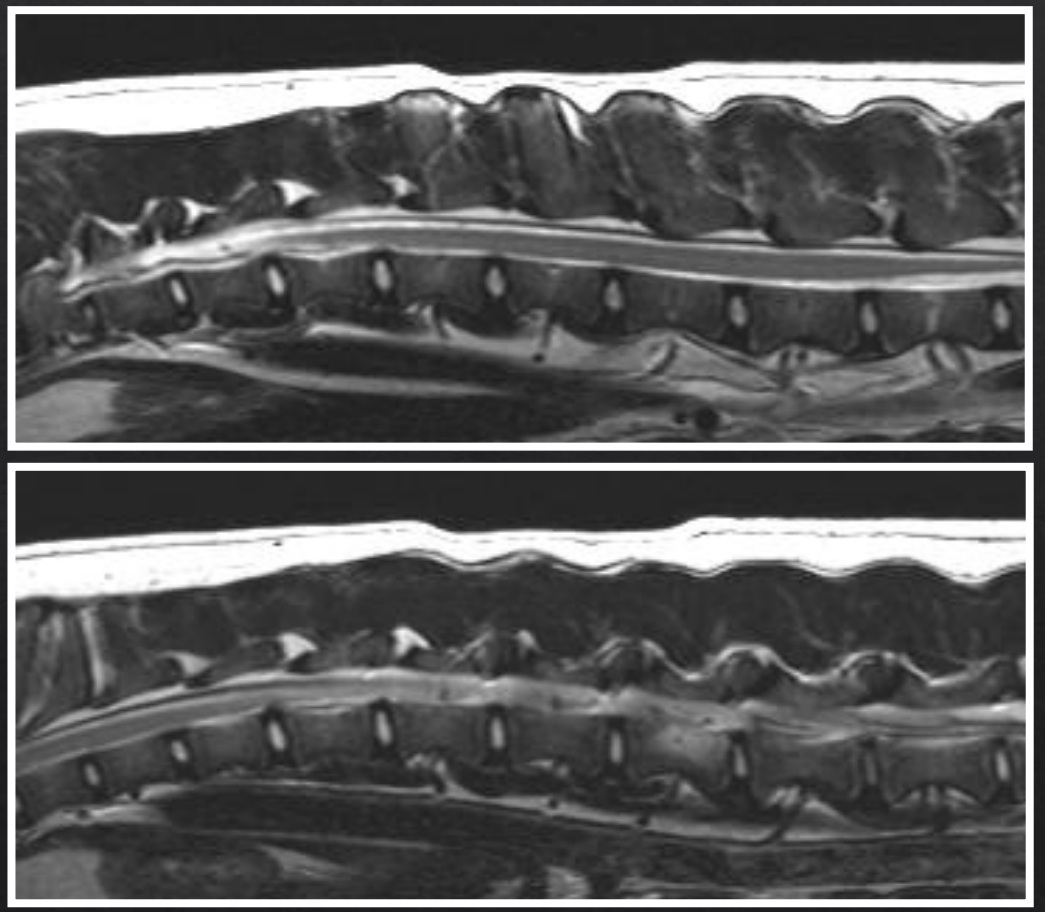


Position

- Appropriate coil
- Target area:

  Close to the coil

  Center of the coil
- Keep the position straight



Significant lesion vs. incidental findings

